

Bryan Smith

Data Scientist | Data Platform & Analytics Engineering

Data Analytics • Business Intelligence • Software Engineering

Azure • Microsoft Fabric • Spark • Python • SQL

Pretoria, South Africa

bryansmith.co.za • linkedin.com/in/bryansmithza • github.com/bryanmehsmith

Summary

Data Scientist and data platform engineer with responsibility across the full lifecycle of production data, analytics, and automation in investment management. Builds Microsoft Fabric Lakehouse pipelines, semantic models, Power BI reporting, enterprise integrations, and secure internal tools. Works directly with stakeholders from requirements and architecture through deployment and support, with a strong focus on testing, least privilege, observability, and root cause diagnosis.

Experience

Data Scientist

Peregrine Capital

Sandton

Nov 2021 – Present

Built and now own core parts of the production data and analytics platform, supporting investment data, reporting, enterprise integrations, and internal operational tooling.

- Engineered a Microsoft Fabric platform spanning Lakehouse, pipelines, PySpark, T-SQL, Direct Lake semantic models, Power BI, paginated reporting, and governed deployment workflows.
- Built ingestion, reconciliation, synchronisation, and delivery flows across portfolio, CRM, finance, web, REST API, SFTP, and file sources.
- Designed and maintained dimensional models, data contracts, database schemas, access controls, and compatible changes across pipelines, semantic models, and reports.
- Introduced event driven ingestion where appropriate, triggering downstream processing when files arrive instead of relying only on estimated delivery schedules.
- Delivered governed investment reporting covering positions, trades, returns, exposure, reconciliation, and investor reporting across multiple business functions.
- Built unsupervised models that flag observations outside expected patterns and engineered features for analysis, including temporal decomposition and derived labels.
- Built secure internal applications, Microsoft Teams bots, and workflow automation using Azure, Microsoft Graph, Adaptive Cards, Entra identities, SAML, and Key Vault.
- Embedded OCR, transcription, LLM extraction, classification, unsupervised modelling, and agent routing into operational workflows, with human review and engineering safeguards. Built a spam detection agent powered by an LLM for inbound messages and a query router agent for tools, workflows, or specialists.
- Diagnosed reporting and data issues end to end by tracing symptoms through reports, semantic models, pipelines, historical records, and source rows.
- Applied automated testing, CI/CD, infrastructure as code, least privilege, observability, deployment gates, and rollback safeguards across platform changes.
- Partner with stakeholders and vendors on requirements, architecture, rollout, production support, platform tradeoffs, and build versus buy decisions. Mentor engineers and review code.

Built ETL pipelines tuned for high performance and data integrations at enterprise scale.

- Built dynamic, reusable ETL pipelines using Ab Initio and Teradata SQL, ingesting from Kafka queues, Postgres, and MySQL to power the App, USSD, and Online channels.
- Redesigned Teradata table structures to balance cost against speed through strategic indexing, structural refinements, and multi value compression (MVC), cutting storage by up to 50%.
- Tuned pipelines to process gigabytes within short batch windows through repeated rounds of indexing, partitioning, and resource management.
- Developed logical and physical data architectures spanning dozens of interconnected tables.
- Managed SQL environment permissions and access controls across production and development layers, minimising risk.
- Earned code review and mentoring responsibilities and delivered production code under senior engineer review.

Education

BSc Computer Science, Part Time 2022 – 2025

University of Witwatersrand, Johannesburg

Relevant coursework: Machine Learning, Parallel Computing, Advanced Analysis of Algorithms, Software Design.

BSc Actuarial and Financial Mathematics 2016 – 2019

University of Pretoria, Pretoria

Relevant coursework: Multivariate Analysis, Stochastic Processes, Time Series Analysis, Financial Engineering.

Actuarial Exams (ASSA)

Actuarial Society of South Africa

Skills

Data Platform: Microsoft Fabric, Lakehouse, Direct Lake, Semantic Models, TMDL, PySpark, T-SQL

Data Processing & Modelling: Spark, Batch Processing, Stream Processing, Airflow, Dimensional Modelling, Data Contracts, Database Design

Databases: SQL Server, Postgres, Teradata, MySQL, Database Administration, Schema Design, Access Controls

Reporting & Domain: Power BI, DAX, Paginated Reports, Investment Data, Positions, Trades, Returns, Exposure, Reconciliation, Investor Reporting

Analytics & Research: Predictive Modelling, Statistical Analysis, Feature Engineering, Factor Regression, Momentum Research

AI & Machine Learning: Azure AI Foundry, LLM Integration, Agentic AI, OCR, Classification, Unsupervised Learning, AI Evaluation, RAG

Cloud & Identity: Azure, Bicep, Container Apps, Functions, Entra ID, OAuth, SAML SSO, Key Vault, Conditional Access

Integration & Applications: REST APIs, FastAPI, SFTP, Salesforce, Microsoft Graph, Teams Bots, Logic Apps

Programming: Python, SQL, TypeScript, JavaScript, React, Node.js, Bash, R (Familiar)

Engineering & Delivery: Docker, Azure DevOps, GitHub Actions, CI/CD, Automated Testing, Observability, Root Cause Analysis

Leadership & Governance: Requirements Analysis, Solution Architecture, Stakeholder Management, Mentoring, Code Review, Build versus Buy

Independent AI Engineering

Built a hybrid coding workflow in OpenCode where hosted frontier models handle orchestration, planning, and review, while Qwen and Gemma models served locally through Ollama take on delegated discovery and bounded implementation tasks. Offloading suitable work to local compute reduces hosted token use.

Languages

Afrikaans (Fluent), English (Fluent)